Proposal Defense
(Ghar Sathi)

Submitted by:

Team

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1. Introduction

1.1 Background

In the contemporary world, digital technology has fundamentally transformed the way services are discovered, accessed and delivered. Online service platforms ranging from ride-sharing and food delivery to freelance marketplaces have demonstrated that connecting service seekers with service providers through a digital intermediary can significantly improve efficiency, reduce costs and enhance user satisfaction. However, the household services sector remains largely undigitized in many developing countries, including Nepal.

Households across Nepal, particularly in urban and semi-urban areas such as Kathmandu Valley, Pokhara, Biratnagar, and Butwal frequently require a diverse range of domestic services. These include routine tasks such as house cleaning, plumbing, electrical repairs, carpentry, painting, appliance servicing, gardening, and caretaking. Despite the high demand for such services, the current mechanism for accessing them is predominantly word-of-mouth referrals, informal local networks or physical searching, all of which are time-consuming, unreliable and often lead to unsatisfactory outcomes.

The rapid growth of smartphones and internet connectivity in Nepal over the past decade has created an environment for digital service platforms with increasing digital literacy among both urban consumers and service workers. There exists a significant opportunity to harness technology to formalize and modernize the household services market. Inspired by global platforms such as Urban Company (India), TaskRabbit (USA), and Handy (UK), Ghar Sathi aims to create a Nepal-centric, culturally appropriate digital solution that caters specifically to the unique demands and challenges of the domestic service sector in Nepal.

Ghar Sathi literally means 'Home Companion' in Nepali which envisions a future where every household can effortlessly find, book, and manage trusted service professionals with just a few clicks, and where skilled workers have a legitimate, visible platform to offer their services to a wider customer base.

1.2 Domain Explanation

The scope of the Ghar Sathi application encompasses the following primary domains:

- Customer Interface: A web-based portal through which registered users can search for service providers by category, location, price range and ratings; view detailed provider profiles, book services, make payments, track booking status; and submit reviews.

- Service Provider Interface: A dedicated portal where skilled workers can register, create and manage their profiles, list the types of services they offer, set pricing and availability, accept or reject booking requests and view their earnings and reviews.

- Administrative Dashboard: A comprehensive management panel for administrators to oversee all platform activities, approve or reject service provider registrations, manage users, resolve complaints, generate reports, and maintain overall platform integrity.

- Notification System: Real-time push notifications for booking confirmations, status updates, payment receipts, and promotional communications.

The application is designed as a responsive web application accessible from desktop and mobile browsers.

1.3 Problem Context

Finding reliable household service providers in Nepal is often difficult and time-consuming. Most people depend on personal contacts, recommendations, or social media to find electricians, plumbers, cleaners, tutors, and other workers. This traditional process is unorganized, making it hard for customers to compare prices, service quality, availability, and trustworthiness. As a result, customers often face delays, uncertainty and limited options.

At the same time, many skilled workers struggle to reach customers and grow their income because they lack a proper platform to promote their services. They mainly rely on local connections and word of mouth, which limits job opportunities. In addition, the absence of a secure system creates issues such as fake bookings, communication problems and lack of transparency in pricing and service quality for both customers and workers.

To solve these problems, Ghar Sathi is proposed as a web-based platform that connects customers with verified service providers in a simple and secure way. The platform allows users to search, compare and book services online, while workers can manage their profiles and reach more customers. Features such as ratings, reviews and real-time notifications help improve trust, convenience, and efficiency in the household service sector.

2. Problem Statement

2.1 Defining the Problem:

In Nepal, finding reliable household service providers is still largely dependent on informal methods like personal contacts, referrals or social media. There is no centralized digital platform where customers can easily search, compare, book and review service providers. This creates difficulties in verifying worker quality, managing bookings, ensuring transparent pricing and maintaining proper communication between customers and workers.

2.2 Who Is Affected:

The problem mainly affects households, working professionals, students and families who require quick and trustworthy household services. Skilled workers and small service providers are also affected because they lack a proper platform to promote their services and reach more customers. Additionally, administrators and businesses in the service sector face challenges in maintaining organized records, trust and accountability within the market.

2.3 Impact:

The absence of a structured platform leads to wasted time, limited service options, communication problems and risks related to unverified workers. Customers may experience poor-quality service, overcharging, delays or security concerns, while service providers struggle to find stable job opportunities and fair competition. Overall, these issues reduce efficiency, transparency and trust in Nepal’s household service industry, slowing the growth of a more organized digital service economy.

3. Objectives

3.1 General Objective

The general objective of the Ghar Sathi project is to design, develop and deploy a comprehensive web-based digital platform that efficiently connects Nepali households with verified, skilled service providers for all categories of domestic work, thereby streamlining the household service ecosystem through technology, improving service quality and promoting digital economic inclusion for informal service workers.

3.2 Specific Objectives

The following specific objectives guide the development and implementation of the Ghar Sathi application:

1. To develop a secure user registration and login system for customers, service providers and administrators.
2. To create a service provider management system where workers can add and manage their profiles, skills, pricing, and availability.
3. To build a service search and booking system that allows customers to search for household services and book appointments easily.
4. To implement a rating and review system that helps customers share feedback and helps users choose trusted service providers.
5. To develop an admin dashboard for managing users, bookings, complaints, and overall platform activities.

4. Requirement Analysis

The requirement analysis for the Ghar Sathi system is conducted through research, discussion with users, and study of similar service platforms. The system requirements are divided into functional and non-functional requirements.

4.1 Functional Requirements:

Functional requirements describe the main features of the system.

User Registration and Login

- Customers, service providers, and admin can register and log in securely.
- Users can manage and update their profiles.
- Password reset functionality will be available.

Service Provider Management

- Service providers can create profiles with skills, experience, pricing, and availability.
- Providers can manage and update their service information.
- Admin can approve or reject provider accounts.

Service Search and Booking

- Customers can search services by category and location.
- Users can view provider profiles and ratings.
- Customers can book services by selecting date and time.
- Providers can accept or reject booking requests.
- Users can view booking history and status.

Rating and Review System

- Customers can rate and review service providers after service completion.
- Ratings and reviews will help users choose trusted providers.

Admin Dashboard

- Admin can manage users, bookings, complaints, and service categories.
- Admin can monitor overall platform activities.

4.2 Non-Functional Requirements

Security

- User information will be protected through secure login and password encryption.
- Role-based access will prevent unauthorized access.

Performance

- The system should load quickly and respond efficiently.
- Search and booking processes should work smoothly.

Reliability

- The platform should operate continuously with minimal downtime.
- Regular backups will help prevent data loss.

Usability

- The system will have a simple and user-friendly interface.
- The platform will be mobile responsive and easy to use.

Scalability

- The system should support future growth and additional users.

Maintainability

- A well-documented, modular codebase following coding standards, with monitoring to enable easy debugging, bug fixes, and feature enhancements.

4.3 Software Requirements:

The software requirements for Ghar Sathi include all the tools and technologies used for designing, developing, testing, and deploying the system. These tools are selected for their simplicity, reliability, and ease of use.

Development Tools

- Visual Studio Code (VS Code): Used as the main code editor for writing and managing the project code.

- Git & GitHub: Used for version control, tracking changes, and storing the project securely.
- Figma: Used for designing user interfaces and creating wireframes and prototypes.

Frontend Technologies

- HTML5: Used to structure web pages.
- CSS3: Used for styling and layout design.
- JavaScript: Used to add interactivity and dynamic features.

Backend Technology

- PHP: Used for server-side development and handling application logic such as user management, bookings, and data processing.

Database

- MySQL: Used to store and manage data such as user information, service listings, bookings, and transactions.

Testing Tools

- Postman: Used for testing APIs and backend functionality.
- Google Chrome Developer Tools: Used for debugging and testing frontend performance.

Deployment Tools

- XAMPP / Apache Server: Used for local development and testing.
- Cloud Hosting / Shared Hosting: Used to deploy the system online for users.

Operating System and Browser

- Operating System: Windows 10/11
- Browser Support: Google Chrome, Mozilla Firefox, Microsoft Edge

5. Methodology:

5.1 Agile Software Development Methodology

Ghar Sathi will be developed using the Agile software development methodology. Agile is an iterative and incremental approach to software development that focuses on flexibility, collaboration, and continuous improvement. It delivers working software in small stages called sprints.

Agile is chosen over traditional models like waterfall because service-based platforms often have changing requirements. As users interact with the system, feedback is collected and features may need to be updated or improved. Agile supports this continuous change effectively.

Agile is suitable for Ghar Sathi because:

- The system includes multiple user types such as customers, service providers, and admin.
- Requirements may evolve during development.
- Continuous feedback and improvement are required.
- The system can be developed and tested in stages.

5.2 Development Phases:

The development process of Ghar Sathi is divided into the following phases:

Phase 1: Requirement Gathering and Planning

In this phase, requirements are collected through interviews and surveys with potential users, including customers and service providers. Secondary research is also conducted by studying similar platforms and relevant literature.

Phase 2: System Design

This phase focuses on designing the system structure. It includes:

- Selecting the technology stack
- Designing the database structure (ERD)
- Planning APIs
- Creating UI/UX wireframes and prototypes

Design tools such as Figma are used to create visual designs of the system. The final design is reviewed with stakeholders before development starts.

Phase 3: Iterative Development (Sprint Cycles)

Development is carried out in short cycles called sprints, usually lasting two weeks. In each sprint, selected features from the backlog are developed, tested, and reviewed.Each sprint includes planning, development, testing, review, and retrospective meetings.

Phase 4: Testing and Quality Assurance

Testing is carried out throughout the development process as well as after completion. It includes:

- Unit testing for individual components
- Integration testing for connected modules
- System testing for the complete application
- Performance testing for speed and reliability

This ensures the system is stable, functional, and meets user requirements.

Phase 5: Deployment

After successful testing, the system is deployed on a cloud platform. A phased rollout approach may be used, starting with a small group of users (for example, in Kathmandu) before full release.

Continuous Integration and Continuous Deployment (CI/CD) tools such as GitHub Actions are used to automate testing and deployment.

Phase 6: Maintenance and Continuous Improvement

After deployment, the system enters the maintenance phase. User feedback is collected and used to improve the platform.

5.3 Justification for Agile Methodology:

Agile is suitable for Ghar Sathi because it allows flexibility and continuous improvement during development. Since user requirements may change over time, Agile ensures the system can adapt easily.

The main advantages of Agile include:

- Flexibility to handle changing requirements
- Early delivery of working features

- Continuous user and stakeholder feedback
- Reduced risk of project failure

Overall, Agile helps ensure that Ghar Sathi is developed in a structured, efficient, and user-focused manner.

6. Use Case Diagram:

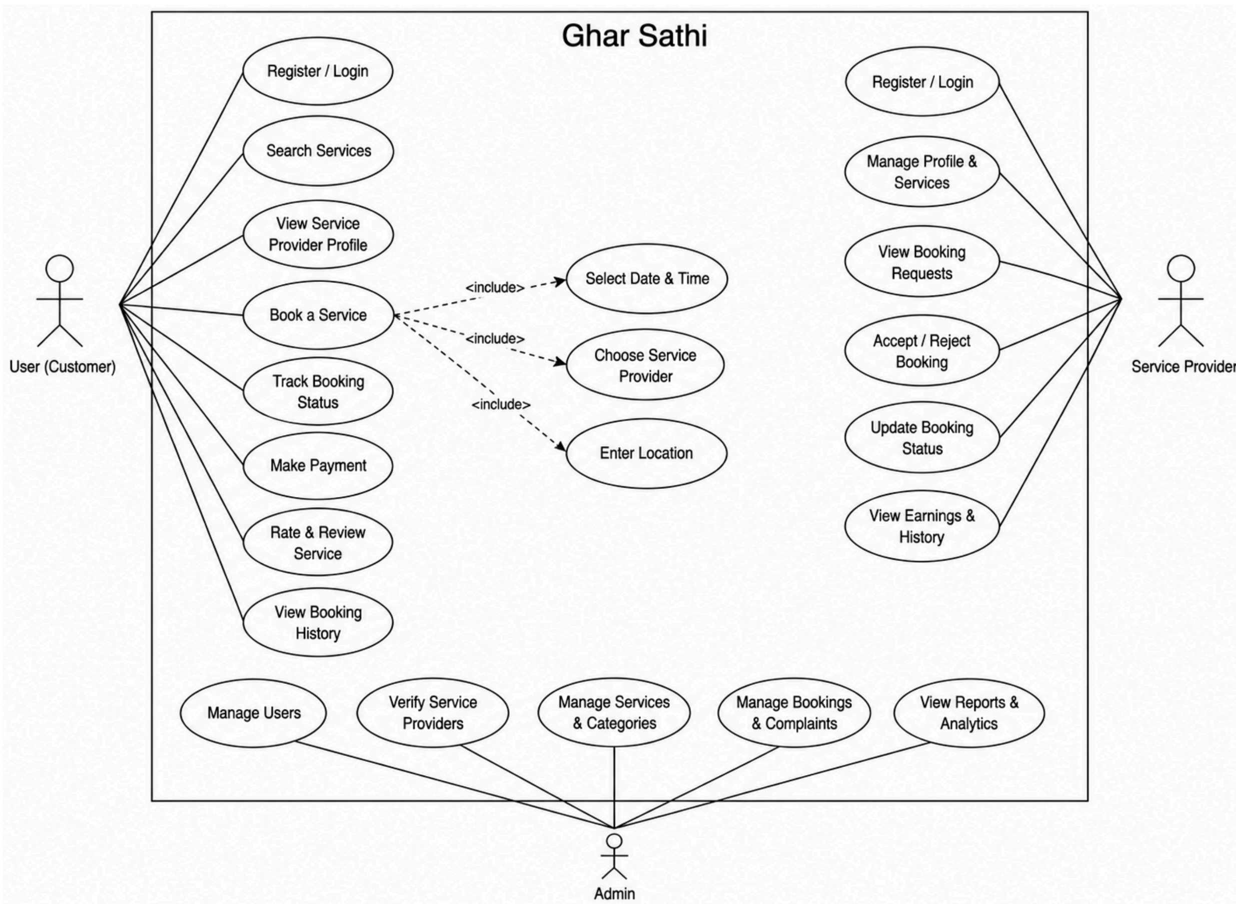

Fig. 6.1 Use case diagram for Ghar Sathi

6.1 Actors

Actor 1: User

The Customer represents any registered household user who accesses the platform to find and engage household service providers. Customers interact with the system to search for services, view provider profiles, make bookings, process payments, track their bookings, and submit reviews and ratings for completed services. The Customer actor also has the ability to file complaints regarding unsatisfactory service experiences.

Actor 2: Service Provider

The Service Provider represents a registered skilled worker who offers domestic services through the Ghar Sathi platform. Service Providers interact with the system to manage their professional profiles, list their services and pricing, set their availability, respond to booking requests (accept or reject), mark bookings as completed, view customer reviews, and manage their earnings and payment history.

Actor 3: Administrator

The Administrator represents authorized platform management personnel who oversee the overall operation of the Ghar Sathi platform. Administrators interact with the system to approve or reject service provider registrations after document verification, manage all user accounts (customers and providers), oversee all bookings and transactions, handle customer complaints, generate operational and financial reports, manage service categories, and ensure the platform adheres to quality standards.

6.2 Use Case Descriptions

Use case of the User:

- Register/Login
- Search services
- View service provider profile
- Book a service
- Track booking status
- Make payment
- Rate and review service
- View booking history

Use case of the Service Provider:

- Register/Login
- Manage profile and services
- View booking requests
- Accept/Reject bookings
- Update booking status
- View earnings and history

Use case of the Admin:

- Manage users
- Verify service providers
- Manage services and categories
- Manage bookings and complaints
- View reports and analytics

7. System Flowchart

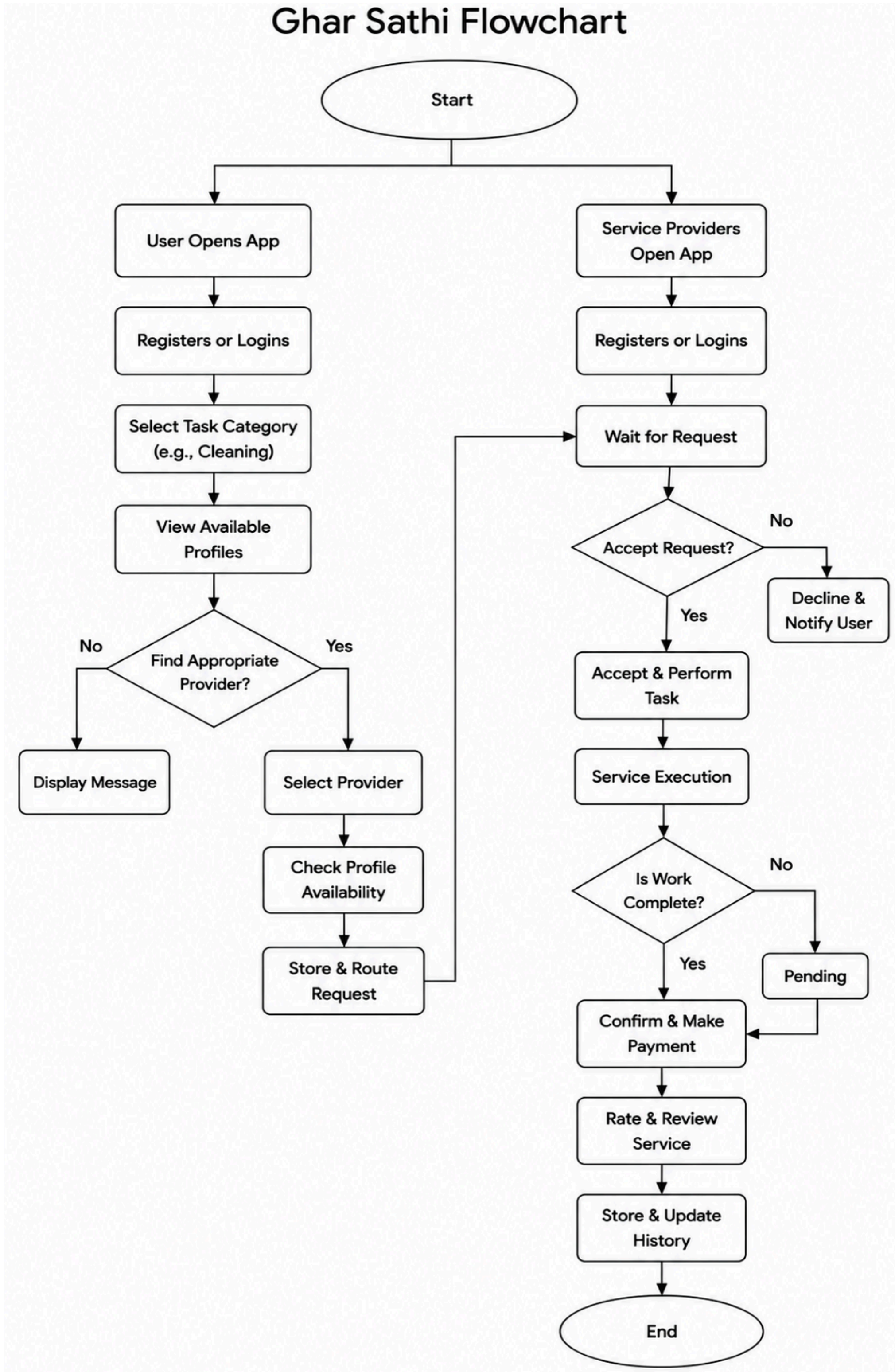

Fig. 7.1 Flowchart for Ghar Sathi

The Ghar Sathi flowchart illustrates the overall working process of the application from the perspective of both customers and service providers. The process begins at the "Start" point, where the system branches into two separate flows: one for users/customers and another for service providers. Both users and providers first open the application and either register for a new account or log into an existing one.

On the customer side, the user selects a service category such as cleaning or plumbing and views the available service provider profiles. The system then checks whether an appropriate provider is available. If no suitable provider is found, a message is displayed to the user. If a provider is available, the customer selects the provider, checks availability, and sends a booking request through the system.

On the service provider side, providers wait for incoming booking requests after logging into the application. Once a request is received, the provider decides whether to accept or reject it. If the request is rejected, the system notifies the customer about the decline. If accepted, the provider proceeds to perform the assigned task.

After accepting the booking, the provider carries out the service execution process. The system then checks whether the work has been completed. If the task is still incomplete, the booking status remains pending until the work is finished. Once the service is successfully completed, the customer confirms the completion and proceeds with payment.

Finally, after payment is completed, the customer can rate and review the service provided. The system then stores and updates the booking history and transaction records. The process concludes at the "End" point, indicating successful completion of the service cycle.

8. Gantt Chart

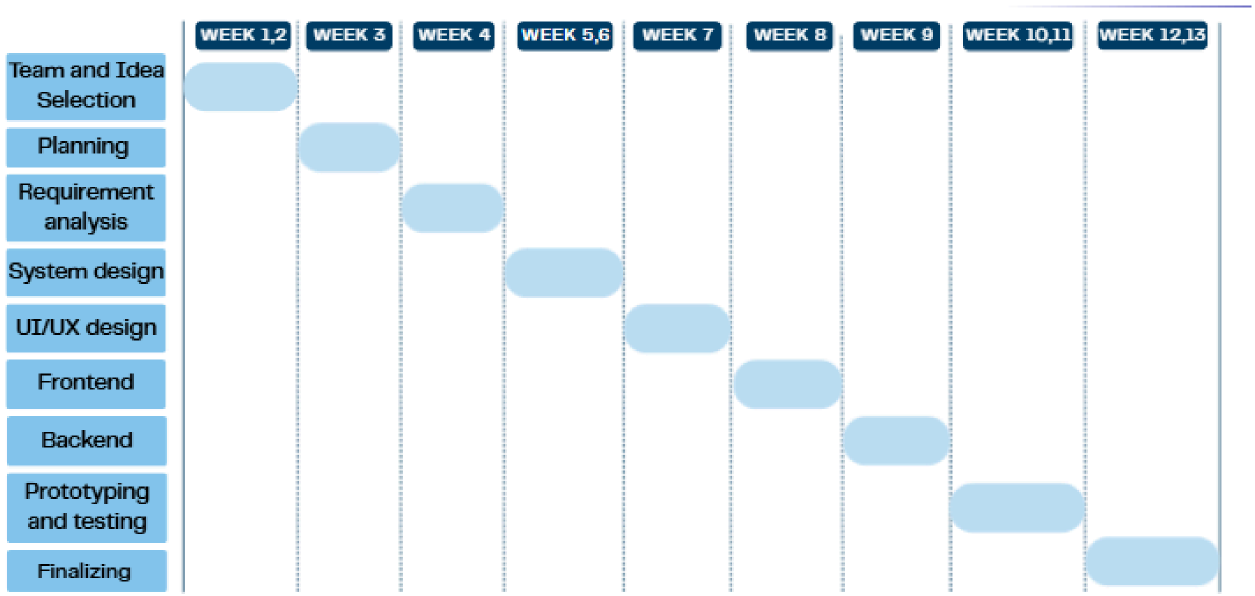

Fig. 8.1 Gantt Chart for Ghar Sathi

This diagram is a Gantt chart that shows the development schedule of the Ghar Sathi project over 13 weeks. The project starts with team and idea selection in Weeks 1–2, followed by planning in Week 3 and requirement analysis in Week 4. System design is completed during Weeks 5–6, while UI/UX design is done in Week 7. Frontend and backend development are carried out in Weeks 8 and 9 respectively. In Weeks 10–11, the system undergoes prototyping and testing to identify and fix issues. Finally, the project is completed during Weeks 12–13 with final improvements.

8.1 Work Division:

Frontend Development

- Nimisha Chapagain
- Aavash Tamang

Backend Development

- Bibhashree Pradhan
- Aayusha Sunuwar

Database Management

- Aayusha Sunuwar

- Nimisha Chapagain
- Aavash Tamang
- Bibhashree Pradhan

UI/UX Design

- Aavash Tamang
- Nimisha Chapagain

Documentation

- Nimisha Chapagain
- Bibhashree Pradhan
- Aayusha Sunuwar
- Aavash Tamang

9. System Architecture

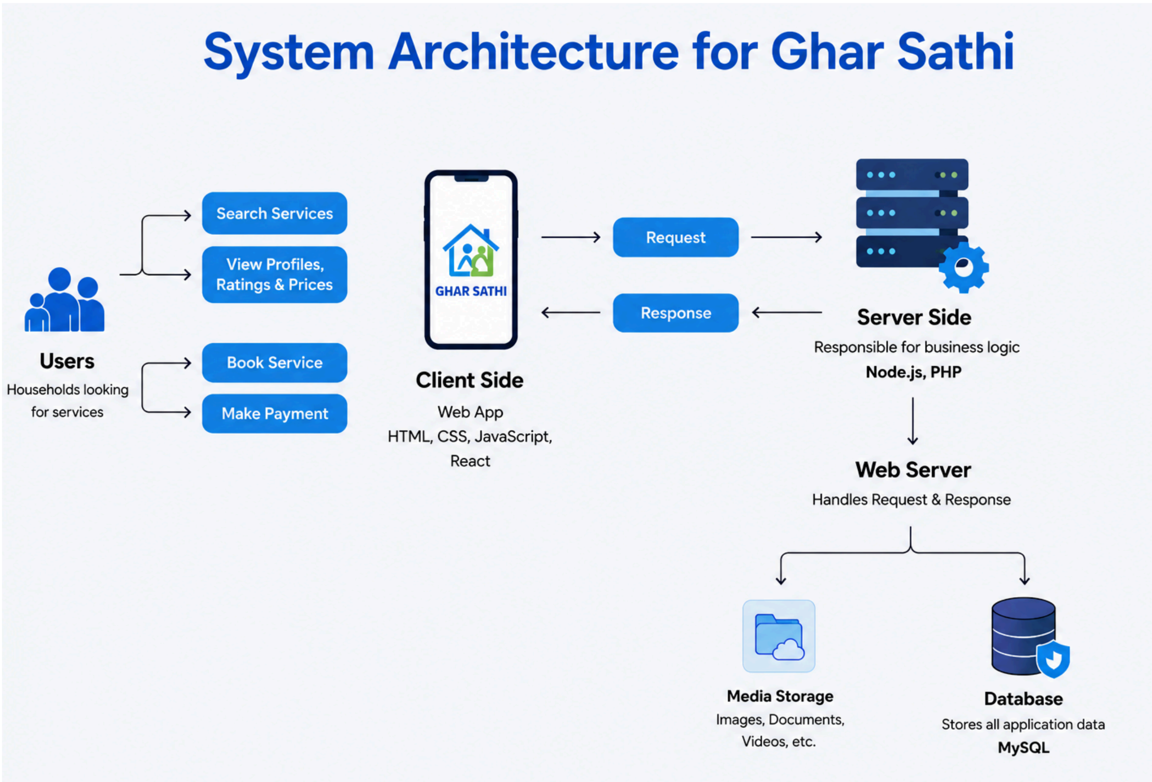

Fig. 9.1 System Architecture for Ghar Sathi

The diagram illustrates the system architecture of Ghar Sathi. Users interact with the application through the client side (frontend), where they can search for services, view provider profiles, check ratings and prices, book services, and make payments. The frontend is built using technologies such as HTML, CSS, JavaScript, and React, which provide an interactive and user-friendly interface.

When a user performs an action, the frontend sends a request to the server side (backend). The backend, developed using Node.js and PHP, handles the business logic of the application, such as processing bookings, managing user data, and handling payments. After processing the request, the server sends a response back to the client side so the user can view the results or confirmation.

The backend is connected to a web server, which manages communication between the client and the server. It also interacts with two important components: media storage and the database. Media storage is used for storing images, documents, and videos, while the MySQL database stores all application data, including user accounts, service details, bookings, reviews, and payment information. Overall, the architecture follows a structured client-server model that ensures efficient communication, data management, and scalability.

10. Expected Outcome

The development of Ghar Sathi is expected to create a more organized, reliable, and convenient way of accessing household services in Nepal. By connecting users with trusted service providers through a digital platform, the system aims to improve both customer experience and service accessibility. The following are the major expected outcomes of the project:

- **Improved Access to Household Services**

  Ghar Sathi will make it easier for users to quickly find trusted service providers for household tasks like cleaning, plumbing, and caretaking through one platform.

- **Significant Time Savings for Households**

  The platform will reduce the time and effort needed to search for workers by providing easy booking and service comparison features.

- **Increased Trust and Service Quality**

  With verified profiles, ratings, and reviews, users will be able to choose reliable workers more confidently, improving overall service quality.

- **Economic Empowerment of Informal Service Workers**

  Service providers will get access to more customers, helping them increase their work opportunities and income through a digital platform.

- **Digital Transformation of the Household Service Ecosystem**

  Ghar Sathi will encourage the use of digital solutions in Nepal's household service sector, making services more organized and accessible.

- **Academic and Technical Contribution**

  The project demonstrates the practical use of web technologies, database management, and software development to solve real-world problems in Nepal.

11. Conclusion

To conclude, Ghar Sathi is a highly relevant, well-designed, and commercially feasible software product for a major challenge faced by Nepali society. The issues related to finding and hiring qualified domestic workers continue to be a critical problem for millions of families and at the same time limit the professional development and business opportunities for thousands of skilled laborers. The described app effectively resolves these problems by developing an interactive digital platform for connecting the demand and supply sides in this segment.

This proposal clearly illustrates how we can effectively apply the principles and concepts learned in the course to develop a complex IT project. Through this process, we were able to identify key problems and define clear goals, conduct detailed requirement analyses in different dimensions, select an appropriate methodology, design a system using several diagrams and flowcharts, and finally develop a realistic plan for the upcoming stages. This experience is invaluable for our professional growth and future career success.

As mentioned above, Ghar Sathi would have several major positive social impacts on society. Firstly, the platform helps to formalize the domestic service market through digitalization, which is vital for promoting economic

12. References

The following references have been consulted in the research, planning, and development of this project proposal:

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